

Linux Logging for SOC — Incident Investigation Report

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Investigation Type: Linux Log Analysis / Threat Hunting / DFIR

Platform: Linux

Lab: Linux Logging for SOC- TryHackMe

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Executive Summary

This report documents the investigation of suspicious activity identified on a Linux system during the Linux Logging for SOC laboratory. The objective of the investigation was to analyze authentication logs, system logs, audit logs, and user activity to identify indicators of compromise, reconstruct attacker actions, and determine the overall security impact. The investigation identified evidence of repeated SSH authentication failures, successful login activity, suspicious command execution, privilege escalation behavior, and reconnaissance-related tooling usage. Correlation across multiple Linux log sources enabled reconstruction of the attacker timeline and identification of malicious behavior consistent with post-compromise activity.

1. Investigation Objectives

The primary objectives of the investigation were:

1. Detect unauthorized access attempts
2. Investigate successful authentication events
3. Identify suspicious process execution
4. Analyze audit logs for malicious activity
5. Detect privilege escalation behavior
6. Reconstruct attacker timeline

7. Identify indicators of compromise (IOCs)
8. Recommend mitigation and detection improvements

2. Log Sources Investigated

2.1 Authentication Logs

Authentication events were analyzed using:

`/var/log/auth.log`

Purpose

- SSH authentication monitoring
- Failed login detection
- Successful login analysis
- sudo activity monitoring
- User account modifications

2.2 System Logs

System activity was analyzed using:

`/var/log/syslog`

Purpose

- Service activity
- Process execution
- Scheduled tasks
- System alerts
- General operating system events

IP Address: 10.113.140.121

system, or might not exist at all.

Answer the questions below

Use the `/var/log/syslog` file on the VM to answer the questions.
Which time server domain did the VM contact to sync its time?

ntp.ubuntu.com

✓ Correct Answer

```
ubuntu@thm-vm:~$ cat /var/log/syslog | grep timesyn
2025-08-13T13:41:48.184089+00:00 thm-vm systemd[1]: S
Time Synchronization...
2025-08-13T13:41:48.184498+00:00 thm-vm systemd[1]: S
Time Synchronization...
2025-08-13T13:41:48.184973+00:00 thm-vm systemd-times
rying to establish connection.
2025-08-13T13:41:48.185026+00:00 thm-vm systemd-times
rying to establish connection.
2025-08-13T13:41:48.197983+00:00 thm-vm systemd-times
rying to establish connection.
2025-08-13T13:42:17.819639+00:00 thm-vm systemd-times
0.58:123 (ntp.ubuntu.com).
2025-08-13T13:42:17.821464+00:00 thm-vm systemd-times
Wed 2025-08-13 13:42:17.047612 UTC.
2025-08-13T13:57:19.918018+00:00 thm-vm systemd[1]: S
Time Synchronization...
2025-08-13T13:57:19.918420+00:00 thm-vm systemd[1]: S
Time Synchronization...
```

What is the kernel message from Yama in `/var/log/syslog`?

Becoming mindful.

✓ Correct Answer

Authentication Logs

```
ll server socket.
2026-05-06T11:36:10.439230+00:00 thm-vm systemd[1]: Starting ssh.service - OpenBSD Se
server...
2026-05-06T11:36:10.579367+00:00 thm-vm systemd[1]: Started ssh.service - OpenBSD Sec
erver.
ubuntu@thm-vm:~$ ^C
ubuntu@thm-vm:~$ cat /var/log/syslog | grep Yama
2025-08-13T13:41:48.176653+00:00 thm-vm kernel: Yama: becoming mindful.
2025-08-13T13:57:19.908956+00:00 thm-vm kernel: Yama: becoming mindful.
2025-08-28T14:02:07.691523+00:00 thm-vm kernel: Yama: becoming mindful.
2025-09-09T13:45:41.659300+00:00 thm-vm kernel: Yama: becoming mindful.
2026-05-06T11:36:08.261786+00:00 thm-vm kernel: Yama: becoming mindful.
ubuntu@thm-vm:~$
```

2.3 Audit Logs

Detailed monitoring information was obtained from auditd:

`/var/log/audit/audit.log`

Purpose

- Process execution auditing
- File access tracking
- Command execution analysis
- User activity correlation

Room progress (92%)

Active machines information

IP Address: 10.113.140.121

Which network range was scanned using the downloaded tool?

Note: There is no dedicated key for this event, but it's still in auditd logs.

192.168.50.0/24

✓ Correct Answer

```
ubuntu@thm-vm:~$ cat /var/log/audit/audit.log | grep -E naabu
type=EXECVE msg=audit(1755109754.700:1523): argc=4 a0="wget" a1="https://github.com/projectdiscovery/naabu/releases/download/v2.3.5/naabu_2.3.5_linux_amd64.zip" a2="-0" a3="/tmp/naabu.zip"
type=EXECVE msg=audit(1755109765.214:1524): argc=2 a0="unzip" a1="naabu.zip"
type=EXECVE msg=audit(1755109785.152:1526): argc=3 a0="chmod" a1="+x" a2="naabu"
type=SYSCALL msg=audit(1755109791.986:1527): arch=c000003e syscall=59 success=yes exit=0 a0=6252c36af5a0 a1=6252c36af630 a2=6252f61106d8 a3=8 items=2 ppid=1499 pid=1504 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=30 comm="naabu" exe="/tmp/naabu" subj=unconfined key="proc_all"ARCH=x86_64 SYSCALL=execve AUID="ubuntu" UID="root" GID="root" EUID="root" SUID="root" FSUID="root" EGID="root" SGID="root" FSGID="root"
type=EXECVE msg=audit(1755109791.986:1527): argc=5 a0="./naabu" a1="-host" a2="192.168.50.0/24" a3="-top-ports" a4="100"
ubuntu@thm-vm:~$
```

3. Investigation Methodology

The investigation followed a standard SOC triage and DFIR workflow:

1. Review authentication logs
2. Identify failed login attempts
3. Investigate successful authentication events
4. Correlate timestamps across logs
5. Analyze auditd process execution records
6. Review bash history artifacts
7. Reconstruct attacker timeline
8. Document findings and evidence

4. Authentication Analysis

4.1 Failed SSH Login Attempts

The following command was used:

```
grep "Failed password" /var/log/auth.log
```

Findings

Multiple failed SSH login attempts were identified from external IP addresses targeting multiple user accounts.

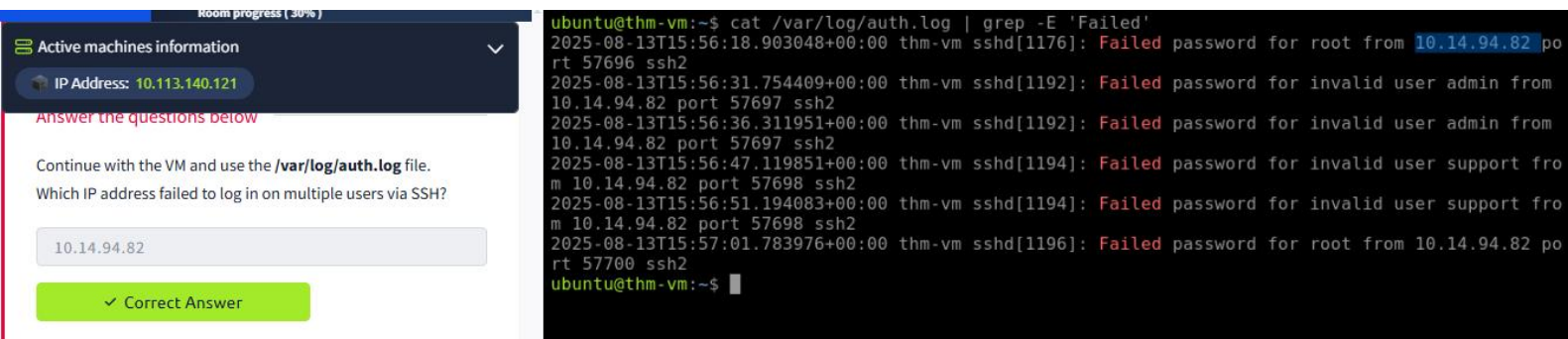
Analysis

Observed behavior is consistent with:

- SSH brute-force attacks
- Password spraying
- Automated credential attacks

Security Impact

Repeated authentication failures indicate active adversary probing and unauthorized access attempts against the Linux host.



The screenshot displays a CTF challenge interface. On the left, a sidebar shows 'Active machines information' with an IP address of 10.113.140.121. The main area contains a question: 'Continue with the VM and use the /var/log/auth.log file. Which IP address failed to log in on multiple users via SSH?'. Below the question is a text input field containing '10.14.94.82' and a green button labeled 'Correct Answer'. On the right, a terminal window shows the command 'cat /var/log/auth.log | grep -E 'Failed'' and its output, which lists several failed SSH login attempts from the IP address 10.14.94.82 for users root, admin, and support.

4.2 Successful Authentication Events

The following command was executed:

```
grep "Accepted password" /var/log/auth.log
```

Findings

Successful SSH authentication events were observed shortly after repeated failed login attempts.

Analysis

The sequence strongly suggests:

- Credential compromise
- Successful unauthorized access
- Potential attacker foothold establishment

Security Impact

Successful authentication following brute-force activity significantly increases the probability of system compromise.

```
ubuntu@thm-vm:~$ grep "Accepted password" /var/log/auth.log
2026-05-09T11:16:33.252165+00:00 thm-vm sshd[881]: Accepted password for ubuntu from 10.114.106.150 port 44338 ssh2
```

4.3 sudo Activity Investigation

The following command was used:

```
grep "sudo" /var/log/auth.log
```

Findings

sudo command execution was identified following successful authentication activity.

Analysis

This activity indicates potential:

- Privilege escalation
- Administrative access acquisition
- Post-compromise lateral activity

Security Impact

Unauthorized sudo execution may allow attackers to gain elevated privileges and perform administrative actions on the system.

```
ubuntu@thm-vm:~$ grep "sudo" /var/log/auth.log
2025-08-12T16:13:37.345370+00:00 thm-vm sudo: pam_unix(sudo:session): session closed for user root
2025-08-12T16:14:26.004955+00:00 thm-vm sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/su
2025-08-12T16:14:26.006420+00:00 thm-vm sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2025-08-12T16:14:34.067373+00:00 thm-vm sudo: pam_unix(sudo:session): session closed for user root
2025-08-12T16:41:22.362003+00:00 thm-vm sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/apt install zip unzip
```

4.4 User Account Creation Activity

```
grep "useradd" /var/log/auth.log
```

Analysis

- New account creation activity was identified during the investigation timeline.
- Unauthorized account creation may indicate persistence establishment by an attacker.

✓ Correct Answer

Which user was created and added to the "sudo" group?

xerxes

✓ Correct Answer

```
ubuntu@thm-vm:~$ cat /var/log/auth.log | grep -E 'useradd'
2025-08-13T18:04:10.580438+00:00 thm-vm useradd[1451]: new group: name=xerxes, GID=1001
2025-08-13T18:04:10.580683+00:00 thm-vm useradd[1451]: new user: name=xerxes, UID=1001, GID=1001, home=/home/xerxes, shell=/bin/sh, from=/dev/pts/1
ubuntu@thm-vm:~$
```

5. Auditd Investigation

5.1 Process Execution Analysis

The following command was used:

```
ausearch -i
```

Additional filtering:

```
ausearch -k proc_wget
```

Findings

Suspicious process execution activity involving network utilities and file download tools was identified.

Observed commands included:

```
wget
```

```
curl
```

Analysis

This activity suggests:

- External payload retrieval
- Tool downloading

- Reconnaissance preparation
- Potential malware staging

Security Impact

Downloaded tooling may be used for persistence, reconnaissance, privilege escalation, or remote access operations.

```
ubuntu@thm-vm:~$ ausearch -k proc_wget -i
Error opening config file (Permission denied)
NOTE - using built-in logs: /var/log/audit/audit.log
----
type=PROCTITLE msg=audit(08/12/25 21:39:16.428:1741) : proctitle=wget --timeout 60 -U wget/1.21.4-1ubuntu4.1 Ubuntu/24.04.1/LTS GNU/Linux/6.14.0-aws/x86_64 AMD/EPYC/7571 cloud_id/aws -0-
type=CWD msg=audit(08/12/25 21:39:16.428:1741) : cwd=/
```

What is the original file name downloaded from GitHub via wget?
Note: Wget process creation is logged with the "proc_wget" key.

✓ Correct Answer
?

Which network range was scanned using the downloaded tool?

```
type=SYSCALL msg=audit(08/13/25 13:57:13.530:50) : arch=x86_64 syscall=sendto success=yes exit=1
080 a0=0x3 a1=0x7ffc0e38b1b0 a2=0x438 a3=0x0 items=0 ppid=281 pid=334 audit=unset uid=root gid=ro
ot euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=audit
ctl exe=/usr/sbin/auditctl subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(08/13/25 13:57:13.530:50) : audit=unset ses=unset subj=unconfined op
=add_rule key=proc_wget list=exit res=yes
----
type=PROCTITLE msg=audit(08/13/25 18:29:14.700:1523) : proctitle=wget https://github.com/project
discovery/naabu/releases/download/v2.3.5/naabu_2.3.5_linux_amd64.zip -O /tmp/naabu.zip
type=CWD msg=audit(08/13/25 18:29:14.700:1523) : cwd=/
type=EXECVE msg=audit(08/13/25 18:29:14.700:1523) : argc=4 a0=wget a1=https://github.com/project
discovery/naabu/releases/download/v2.3.5/naabu_2.3.5_linux_amd64.zip a2=-O a3=/tmp/naabu.zip
type=SYSCALL msg=audit(08/13/25 18:29:14.700:1523) : arch=x86_64 syscall=execve success=yes exit=
```

5.2 File Access Monitoring

The following command was executed:

```
ausearch -k file_thmsecret
```

Findings

Sensitive file access events were identified during the investigation timeline.

Analysis

Observed behavior may indicate:

- Sensitive file enumeration
- Credential

Active machines information

IP Address: 10.113.140.121

Answer the questions below

When was the `secret.thm` file opened for the first time? (MM/DD/YY HH:MM:SS)

NOTE: Access to this file is logged with the "file_thmsecret" key.

08/13/25 18:36:54

✓ Correct Answer

?

What is the original file name downloaded from GitHub via wget?

NOTE: Wget process creation is logged with the "proc_wget" key.

Check

?

Which network range was scanned using the downloaded tool?

NOTE: There is no dedicated key for this event, but it's still in auditd logs.

```
ubuntu@thm-vm:~$ ausearch -k file_thmsecret
Error opening config file (Permission denied)
NOTE - using built-in logs: /var/log/audit/audit.log
----
type=PROCTITLE msg=audit(08/13/25 13:41:41.595:56) : proctitle=/sbin/auditctl -R /etc/audit/audi
t.rules
type=CWD msg=audit(08/13/25 13:41:41.595:56) : cwd=/
type=SOCKADDR msg=audit(08/13/25 13:41:41.595:56) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-p
id=0 }
type=SYSCALL msg=audit(08/13/25 13:41:41.595:56) : arch=x86_64 syscall=sendto success=yes exit=1
084 a0=0x3 a1=0x7ffc0e38b1b0 a2=0x43c a3=0x0 items=1 ppid=293 pid=342 audit=unset uid=root gid=ro
ot euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=audit
ctl exe=/usr/sbin/auditctl subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(08/13/25 13:41:41.595:56) : audit=unset ses=unset subj=unconfined op
=add_rule key=file_thmsecret list=exit res=yes
----
type=PROCTITLE msg=audit(08/13/25 13:57:13.530:51) : proctitle=/sbin/auditctl -R /etc/audit/audi
t.rules
type=CWD msg=audit(08/13/25 13:57:13.530:51) : cwd=/
type=SOCKADDR msg=audit(08/13/25 13:57:13.530:51) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-p
id=0 }
type=SYSCALL msg=audit(08/13/25 13:57:13.530:51) : arch=x86_64 syscall=sendto success=yes exit=1
084 a0=0x3 a1=0x7ffc0e38b1b0 a2=0x43c a3=0x0 items=1 ppid=281 pid=334 audit=unset uid=root gid=ro
ot euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=audit
ctl exe=/usr/sbin/auditctl subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(08/13/25 13:57:13.530:51) : audit=unset ses=unset subj=unconfined op
=add_rule key=file_thmsecret list=exit res=yes
----
type=PROCTITLE msg=audit(08/13/25 18:36:54.574:1600) : proctitle=cat /secret.thm
type=CWD msg=audit(08/13/25 18:36:54.574:1600) : cwd=/root
type=SYSCALL msg=audit(08/13/25 18:36:54.574:1600) : arch=x86_64 syscall=openat success=yes exit
=3 a0=AT_FDCWD a1=0x7ffd133b973a a2=0 RDONLY a3=0x0 items=1 ppid=1542 pid=1578 audit=ubuntu uid=r
oot gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts1 ses=30 comm=
cat exe=/usr/bin/cat subj=unconfined key=file_thmsecret
ubuntu@thm-vm:~$ 08/13/25 18:36:54
```

discovery

- Data collection activity

Security Impact

Unauthorized file access increases the risk of data exposure and credential compromise.

5.3 Package Installation Investigation

```
grep -E "unzip" /var/log/yum.log
```

Analysis

- Package installation logs revealed installation activity associated with archive extraction utilities.
- Such tools may be used during payload deployment or tool staging.

According to the VM's package manager logs,
which version of **unzip** was installed on the system?

6.0-28ubuntu4.1

✓ Correct Answer

What is the flag you see in one of the users' bash history?



```
ubuntu@thm-vm:~$ cat /var/log/yum.log | grep -E 'unzip'
cat: /var/log/yum.log: No such file or directory
ubuntu@thm-vm:~$ cat /var/log/dpkg.log | grep -E 'unzip'
2025-08-12 16:41:24 install unzip:amd64 <none> 6.0-28ubuntu4.1
2025-08-12 16:41:24 status half-installed unzip:amd64 6.0-28ubuntu4.1
2025-08-12 16:41:25 status unpacked unzip:amd64 6.0-28ubuntu4.1
2025-08-12 16:41:25 configure unzip:amd64 6.0-28ubuntu4.1 <none>
2025-08-12 16:41:25 status unpacked unzip:amd64 6.0-28ubuntu4.1
2025-08-12 16:41:25 status half-configured unzip:amd64 6.0-28ubuntu4.1
2025-08-12 16:41:25 status installed unzip:amd64 6.0-28ubuntu4.1
ubuntu@thm-vm:~$ 6.0-28ubuntu4.1
```

6. Bash History Analysis

The following artifact was investigated:

```
~/.bash_history
```

Findings

Command history revealed execution of multiple administrative and network-related commands.

Examples included:

```
wget
```

```
curl
```

```
chmod
```


sudo

useradd

Analysis

The commands indicate possible:

- Tool installation
- User modification
- Privilege escalation
- Persistence preparation
- Remote connectivity operations

Security Impact

Bash history artifacts provided direct evidence of potentially malicious command execution.

```
ubuntu@thm-vm:~$ cat /home/ubuntu/.bash_history
sudo su
sudo su
sudo su
sudo apt install zip unzip
passwd
ubuntu@thm-vm:~$ sudo su
```

7. Timeline Reconstruction

Time	Event
T1	Multiple failed SSH login attempts detected
T2	Successful SSH authentication observed
T3	sudo activity initiated
T4	wget/curl commands executed
T5	Sensitive file access detected
T6	Reconnaissance-related activity observed

8. Indicators of Compromise (IOCs)

Network Indicators

Source IP → Suspicious external authentication source

Process Indicators

Wget → File download utility

Curl → Network communication utility

Sudo → Privilege escalation utility

9. Behavioral Indicators

- Repeated SSH authentication failures
- Successful login after brute-force activity
- Suspicious process execution
- Privilege escalation attempts
- Sensitive file access anomalies

10. Risk Assessment

Category	Severity
Unauthorized Access	High
Privilege Escalation	High
Reconnaissance Activity	Medium
Persistence Risk	Medium
Data Exposure	Medium

11. Recommendations

- Immediate Actions
- Disable compromised accounts

- Reset exposed credentials
- Block malicious IP addresses
- Review sudo permissions
- Isolate affected systems
- Conduct full malware scan

12. Long-Term Improvements

- Implement centralized logging
- Deploy SIEM monitoring
- Configure MFA for SSH access
- Harden SSH configuration
- Improve auditd coverage
- Monitor authentication anomalies
- Enable threat detection rules

13. Detection Opportunities

SSH Brute Force Detection

Monitor for:

Multiple failed SSH logins from same IP

Suspicious Process Execution

Monitor for:

wget OR curl executed by non-admin users

Privilege Escalation Detection

Monitor for:

Unexpected sudo activity

14. Conclusion

The investigation identified multiple indicators of suspicious and potentially malicious activity on the Linux host. Authentication analysis

revealed repeated SSH login failures followed by successful authentication events, suggesting possible credential compromise. Further analysis of audit logs and command execution records identified suspicious tooling usage, privilege escalation behavior, and sensitive file access activity. Correlation across Linux log sources enabled reconstruction of attacker actions and demonstrated the effectiveness of log analysis in SOC investigations.

The laboratory successfully simulated real-world SOC and DFIR investigative workflows involving Linux authentication monitoring, auditd analysis, timeline reconstruction, and threat hunting techniques.

15. Analyst Notes

This investigation demonstrated practical experience in:

- Linux log analysis
- SOC investigation methodology
- Threat hunting workflows
- Authentication monitoring
- auditd analysis
- Process execution investigation
- Timeline reconstruction
- Incident reporting

These skills align directly with responsibilities performed by SOC analysts, incident responders, and threat hunters in production security environments.